Hand written Gradient Descent Date 28/02/2026

Group 1

Part 1 (Iteration 1) by Michael Nwuju

$m = -1$, $\hat{y} = mx + b$

$b = 1$ $\alpha = 0.1$

x	y	$\hat{y}$	
1	3	0	$= -1(1) + 1 = -1 + 1 = 0$
3	6	-2	$= -1(3) + 1 = -3 + 1 = -2$

$Loss(L) = \hat{y} - y$

$L_1 = \hat{y}_1 - y_1 = 0 - 3 = -3$

$L_2 = \hat{y}_2 - y_2 = -2 - 6 = -8$

Mean Squared Error $(J(m,b)) = 1/n \sum_{i=1}^{n} L_i^2$

where n = number of data points

for our case, $n = 2$

$= \frac{1}{2} \times (L_1^2 + L_2^2) = \frac{1}{2}((-3)^2 + (-8)^2)$

$= 1/2(9 + 64) = \frac{73}{2} = 36.5$

Date / /

$J(m,b) = 36.5$, $m = -1$, $b = 1$

$L_1 = -3$, $L_2 = -8$, $x_1 = 1$, $y_1 = 3$

$x_2 = 3$, $y_2 = 6$

Now we know J, we want to minimize by differentiating J with respect to m & b.

We will use the chain rule for this

$$\frac{dJ}{dm} = \frac{dJ}{du} \cdot \frac{du}{dm}$$

(u) will be the Loss function $\rightarrow \hat{y} - y$

Remember, $\hat{y} = mx + b$

$\therefore \quad u = mx + b - y$, $J(m,b) = \frac{1}{2} \Sigma u^2$

So, $\frac{dJ}{du} = \left(\frac{1}{2} \times 2\right) \Sigma u^{(2-1)} = \Sigma u$

Then, $\frac{du}{dm} = u = mx + b - y$, $(1+1)\, m^{(1-1)} x + b^{(1-1)} - y^{(1-1)}$

$\frac{du}{dm} = 1(x) = x$

$$\therefore \quad \frac{dJ}{dm} = \sum_{i=1}^{2} ux$$

Date / /

$$\frac{dJ}{db} = \frac{dJ}{du} \cdot \frac{du}{db}$$

We already know dJ/du so we focus on du/db

$$\frac{du}{db} = 1$$

$$\therefore \frac{dJ}{db} = \sum_{i=1}^{2} u$$

Back to the Math →

$$dJ/dm = L_1 x_1 + L_2 x_2$$
$$= (-3)(1) + (-8)(3)$$
$$= -3 - 24$$
$$= -27$$

$$dJ/db = L_1 + L_2$$
$$= -3 + (-8)$$
$$= -11$$

Date / /

$dJ/dm = -27$, $dJ/db = -11$, $\alpha = 0.1$

$m_{old} = -1$, $b_{old} = 1$

Gradient Descent =

$m_{new} = m - \alpha(dJ/dm)$

$b_{new} = b - \alpha(dJ/db)$

$= m_{new} = -1 - (0.1)(-27)$

$= -1 + (0.1 \times 27)$

$= -1 + 2.7$

$= 1.7$

$b_{new} = 1 - (0.1)(-11)$

$= 1 + (0.1 \times 11)$

$= 1 + 1.1$

$= 2.1$

First Iteration Complete, new values of m & b are -

$m = 1.7$, $b = 2.1$

Alliane UMUTONIWASE

MEMO No.

DATE / /

Iteration 2

$\hat{y} = mx + b$

$m_1 = 1.7$ $\quad x_1 = 1$ $\quad y_1 = 3$

$b = 2.1$ $\quad x_2 = 3$ $\quad y_2 = 6$

$\hat{y}_1 = 1.7(1) + 2.1 = 3.8$

$\hat{y}_2 = 1.7(3) + 2.1 = 5.1 + 2.1$

$= 7.2$

Loss $(L) = \hat{y} - y$

$L_1 = 3.8 - 3 = \underline{0.8}$

$L_2 = 7.2 - 6 = \underline{1.2}$

Let's use the chain rule;

$\hat{y} = mx + b$

$J = \frac{1}{2} \sum (\hat{y} - y)^2$

let;

$u = \hat{y} - y = mx + b - y$

$\frac{dJ}{dm} = \frac{dJ}{du} \cdot \frac{du}{dm}$

Scanned with CamScanner

MEMO No.

DATE / /

Outer derivative;

$$\frac{dJ}{du} = u$$

Inner derivative;

$$\frac{du}{dm} = x$$

Let's multiply the chain rule;

$$\frac{dJ}{dm} = u \cdot x$$

$$\frac{dJ}{dm} = (mx + b - y)\,x$$

For all the points;

$$\frac{dJ}{dm} = \sum (mx_i + b - y_i)\,x_i$$

$$\frac{dJ}{db} = \sum (mx_i + b - y_i)$$

$$\frac{dJ}{dm} = L_1 x_1 + L_2 x_2$$

$$= (0.8)(1) + (1.2)(3)$$

$$= 0.8 + 3.6$$

$$= 4.4$$

Scanned with CamScanner

MEMO No.

DATE / /

$$\frac{dJ}{db} = L_1 + L_2$$

$$db = 0.8 + 1.2$$

$$= 2$$

Let use Gradient Descent;

$$m_{new} = m - \alpha (dJ/dm)$$

$$b_{new} = b - \alpha (dJ/db)$$

$$m_{new} = 1.7 - (0.1)(4.4)$$

$$= \underline{\underline{1.26}}$$

$$b_{new} = 2.1 - (0.1)(2)$$

$$= \cancel{2.1} \; 2.1 - 0.2$$

$$= \underline{\underline{1.9}}$$

Scanned with CamScanner

3rd Iteration by Hamse jama

$\hat{y} = mx + b$ $\quad m = 1.26 \quad b = 1.9 \quad x_1 = 1 \quad x_2 = 3 \quad y_1 = 3 \quad y_2 = 6$

$\hat{y}_1 = 1.26(1) + 1.9 = 1.26 + 1.9 = 3.16$ //

Loss $L_1 = \hat{y}_1 - y_1 = 3.16 - 3 = 0.16$ //

$\hat{y}_2 = 1.26(3) + 1.9 = 3.78 + 1.9 = 5.68$ //

$L_2 = \hat{y}_2 - y_2 = 5.68 - 6 = -0.32$ //

Gradient desc.nt

lets use the Chain rule

$\hat{y} = mx + b$, $J = 1/2 \sum u^2$ $\quad$ let $u = \hat{y} - y = mx + b - y$

Outer derivative $= \frac{dJ}{du} = \frac{d}{du}\left(\frac{1}{2}u^2\right) = u$

Inner derivative for m $= \frac{du}{dm} = x$, $\quad$ inner derivative for b $= \frac{du}{db} = 1$

lets multipy the chain rule

$\frac{dJ}{dm} = u \cdot x = (mx + b - y)x$, $\quad \frac{dJ}{db} = \sum (mx_i + b - y_i)$

~~$x = 1 : \hat{y}_1 =$~~

Apply formulas

$\frac{dJ}{dm} = \sum (u_i \cdot x_i) = (0.16) + (-0.32 \cdot 3) = -0.8$

$\frac{dJ}{dm} = \sum u_i = 0.16 + (-0.32) = -0.16$

$m_{new} = m - \alpha\left(\frac{dJ}{dm}\right) = 1.26 - 0.1(-0.8)$

$= 1.34$ //

$b_{new} = b - \alpha\left(\frac{dJ}{dm}\right) = 1.9 - 0.1(-0.16)$

$= 1.916$ //

4th Heration by Divine Hou

$\hat{y} = mx + b$

$m = 1.34$

$b = 1.916$

$x_1 = 1 \quad x_2 = 3 \quad y_1 = 3 \quad y_2 = 6$

Predictions and losses

$\hat{y}_1 = 1.34(1) + 1.916$

$= 1.34 + 1.916 \Rightarrow 3.256$

Loss $L_1 = \hat{y}_1 - y_1$

$3.256 - 3 = 0.256$

$\hat{y}_2 = 1.34(3) + 1.916 = 4.02 + 1.916$

$= 5.936$

Loss $L_2 = \hat{y}_2 - y_2$

$= 5.936 - 6$

$= -0.064$

Gradient derivation (using chain rule)

$\hat{y} = mx + b \qquad J = \frac{1}{2}\sum u^2$

Let $u = \hat{y} - y = mx + b - y$

$$\frac{dJ}{du} = \frac{d}{du}\left(\frac{1}{2}u^2\right) = u$$

$$\frac{du}{dm} = x_i \quad \Big| \quad \frac{du}{db} = 1$$

So multiple the chain rule

$$\frac{\partial J}{\partial m} = u \cdot x = (mx + b - y)\,x \qquad \frac{\partial J}{\partial b} = \sum(mx_i + b - y_i)$$

plying the formulas:

$$\frac{\partial J}{\partial m} = \cancel{\sum(u_i \cdot x_i) = (0.256(1))}$$

$$\sum(u_i \cdot x_i) = (0.256(1) + (-0.064)(3)$$

$= 0.256 - 0.192 = 0.064$

$$\frac{\partial J}{\partial b} = \sum u_i = 0.256 + (-0.064) = 0.192$$

Update m and b

$$m_{new} = m - \alpha\left(\frac{\partial J}{\partial m}\right) = 1.34 - 0.1(0.064$$

$$= 1.3336$$

$$b_{new} = b - \alpha\left(\frac{\partial J}{\partial b}\right) = 1.916 - 0.1(0.192)$$

$$= 1.8968$$

$\therefore m_{new} = 1.3336$

$b_{new} = 1.8968$